

Modeling of Annual Row Crop Yield via Generalized Linear Mixed Models

Jacob M. Knodle
North Dakota State University
Department of Statistics

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Abstract

This project investigates the relationship between various agriculture managements and yield outcomes. Data for the project comes from the Main Cropping System Experiment (MCSE), the primary long-term research experiment performed at the Kellogg Biological Station, directed by a team at Michigan State University. The MCSE focuses on studying the interactions between soil chemistry gradients, crop types, and land use practices. The primary goal of our analysis is to model crop yield as a function of treatment type, chemical input, and environmental factors while addressing the non-normal nature of crop yields. This study of the experimental data falls under the framework of the generalized linear model approach, specifically examining relationships under the assumptions of the Gamma and Quasi models. Our final analysis aims to quantify how land management intensity impacts yield outcomes and soil chemistry for future planting seasons so that more effective and sustainable crop management methods can be used in the agricultural industry.

1 Introduction

1.1 Background

Sustainability, both of the human population and the environment in which it resides, remains a large and persistent issue. World population is projected to exceed 10 billion in the next 50 years, requiring a global food production increase by an estimated 70% [1]. In order to meet these rising needs, the search for improved agricultural practices has increased. The use of fertilizer, herbicide, and traditional land tilling has been shown to consistently increase row crop yields compared to no till and reduced fertilizer/herbicide treatments, though often at the cost of damaging soil composition and native plant, animal, and insect species [2][3]. Quantifying this tradeoff, along with fine tuning the methods used for improvement in

crop yields is still a topic for research and discussion, one of which we shall explore further in this paper.

Traditional linear regression methods are commonly used to analyze the relationships between these treatments and their effect on crop yield. However, the assumptions, particularly that of residual structure, are often severely violated in such contexts. Luckily, we have extensions of these linear models that can help deal with these violations. We will utilize methods of generalized linear and mixed-effect models to assess the unique characteristics of yield analysis [4]. We will fit these models on data taken from annual crop plots as a part of the Main Cropping System Experiment (MCSE) at the Kellogg Biological Station and interpret the results.

1.2 Research Objectives

Through our analysis of the MCSE, we sought to evaluate the relationship between soil nutrient composition and row crop yield. A natural extension of this objective is to determine the most effective land management, fertilizer, and herbicide treatments for increasing crop yield and to determine the magnitude of such treatment. Finally, we will assess the suitability of generalized linear models and mixed-effects models for the purpose of examining the aforementioned relationships.

2 Materials and Methods

2.1 Study Area

Data was collected across 6 replicate plots for 4 different land management treatments (amounting to 24 total plots). These plots were located at the Kellogg Biological Station in Southwest Michigan. Each of the 1 ha plots has been assigned to one of the four treatments since 1989 and are still active as of the writing of this paper (2026).

2.2 Treatments

The Main Cropping System Experiment focuses on three primary areas of treatment: land management technique, fertilizer application, and herbicide application. Land management

techniques in the study are divided into four distinct treatments, denoted as Conventional (T1), No-till (T2), Reduced Input (T3), and Biologically Based (T4). Conventional and No-till treatments received standard levels of fertilizer and herbicide input along with being chisel plowed and untilled respectively. The Reduced Input treatment received banded herbicide and nitrogen based fertilizer along with post-planting cultivation. The Biologically Based treatment received no fertilizer or herbicide application, but was rotary-hoed to control weeds.

In addition to controlling the land management treatments, the rate of fertilizer and herbicide application (measured in kg/ha) were recorded. Fertilizer application measurements were separated into the application rates of three primary fertilizer nutrients: Nitrogen, Phosphorus, and Potassium. Herbicide application rate was measured by general rate of application, while it should be noted that there is discussion as to the variation in effectiveness of different herbicide strains [5][6].

2.3 Response Variable

The primary response variable associated with the study is the measured yield of each type of row crop (Corn, Soybeans, and Winter Wheat) in bu/acre. This is measured for each plot during harvest.

3 Exploratory Data Analysis

3.1 Response Variable

We will begin by examining the empirical distribution of the response variable (yield) both on the original and log scales. By examining the original scale, we can identify a few key

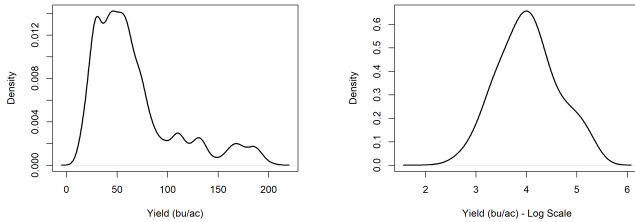


Figure 1: Distribution of raw yield values (left) and log-transformed yield values (right).

characteristics of the response variable. We see that the distribution of yield is right skewed and strictly positive. This suggests the potential use of a log transformation for normal approximation, which can be seen above to significantly improve the normality of the response variable. One additional characteristic that we will examine is the relationship between the observed mean value and the variance.

After examining the mean-variance relationship shown in figure 2, we notice that the variance appears to increase exponentially with the mean. Informal analysis suggests that the variance may be increasing at a rate of $\text{Var}(\mu) = \phi\mu^2$. We will

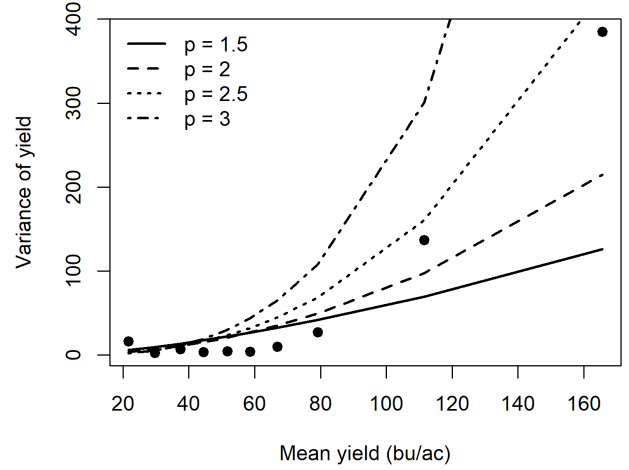


Figure 2: Relationship between the observed mean and variance of yield across grouped MCSE plots via binned mean-variance graph. Variance functions of the form $\text{Var}(\mu) = \phi\mu^p$ are overlaid for several values of p .

examine later if this potential relationship is valid. For now we will note that our analysis of the response variable suggests the use of a log-transformation for yield and a variance function that is scaled by μ^p .

3.2 Explanatory Variables

Our predictor variables can be divided into two groups: land management technique and chemical input levels. Land management technique primarily entails the level of top soil disturbance and cover crop usage, while chemical input levels deal with the rate of herbicide and fertilizer application. However, there is some natural correlation between these two types of variables, as certain land management treatments are denoted as low chemical input.

As previously defined, land management techniques are grouped into four separate treatments in a natural order of descending soil disturbance, denoted as T1, T2, T3, and T4. We can visually summarize their impact on yield in Figure 3.

While land management treatment is a qualitative factor, chemical inputs (fertilizer and herbicide application rates), are quantitative measures. Their relationship with yield can be visualized by grouping chemical input rates into ranges and graphing their distributions, as shown in Figure 4.

Figure 4 suggests that there may be a higher order interaction involved with nutrient rates via fertilizer application. Our plots suggest that there may be an optimal range of application for each fertilizer nutrient that corresponds with the needs of the given crop. Too little application may leave the crop nutrient deficient, while too much may over-saturate the plant. One may deduce that, due to the varying nutrient needs of different crops, there may be a significant interaction between crop species and fertilizer rates, causing different optimal levels of fertilizer for each row crop [7]. Herbicide application rate, on the other hand, appears to adhere more closely to a positive lin-

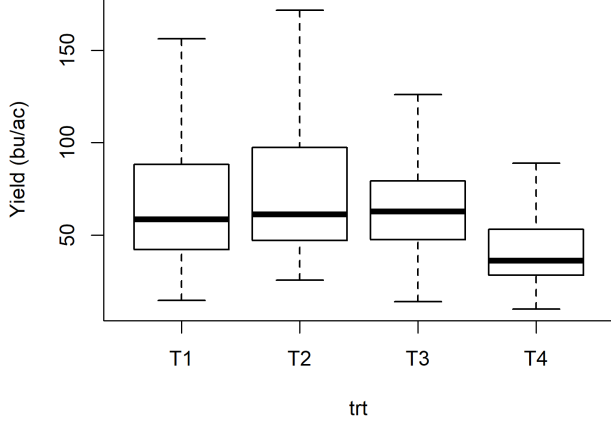


Figure 3: Distribution of row crop yields (bu/ac) by land management treatment - T1, T2, T3, T4.

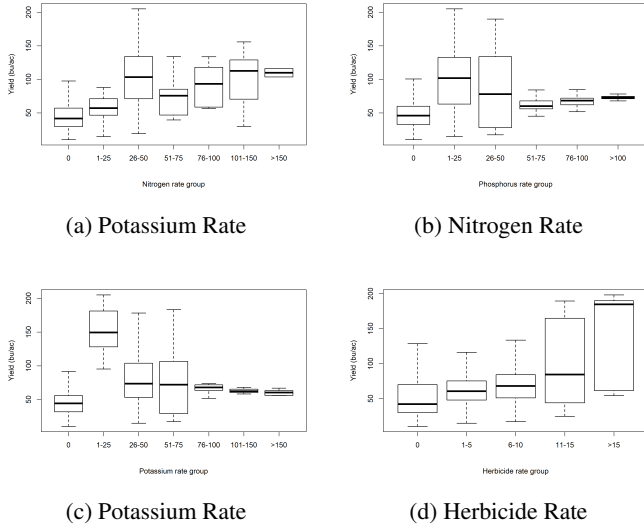


Figure 4: Distribution of crop yield across grouped fertilizer and herbicide application rates.

ear trend of the first order, demonstrating that increased herbicide use improves yield rate rather consistently. We may now choose to move forward with the assumption that fertilizer application may have the strongest interaction with crop species out of the explanatory variables that we have examined.

4 Statistical Methods

Model specification naturally becomes one of the most important factors in our analysis due to the deviations from standard normality assumptions that are seen in the response. While methods of ordinary linear regression have become a powerful tool in analysis of relationships, both univariate and multivariate, since it was first introduced by Galton (1886), we will extend our knowledge of linear models to a more flexible approach with the use of Generalized Linear Models and

Quasi-Generalized Linear Models [4].

Before proceeding, we will make the important note that crop yields are highly influenced by the complex factors associated with each individual planting season. The extremely complex and unique nature of each planting season, primarily due to largely varying environmental factors, is difficult to quantify and a nuisance for generalizing our results to future seasons. Therefore, we will utilize the methods of mixed-effect models described by Faraway (2016) with the use of planting season as a random effect within the scope of each of our models.

Based on the information gathered from our exploratory analysis, we will choose two primary candidate models: a log-normal linear regression model and a gamma generalized linear model. We will examine how the use of each of these models helps to meet the standard assumptions required for reliable inference, utilizing the standard Gaussian ordinary linear regression model as our basis for comparison. In addition, the use of quasi-generalized linear models may be considered if it is determined that our data requires a more flexible modeling approach. The reader should note that for the purpose of model selection and assumption diagnostics, we will utilize only the simple additive models to help address issues of model complexity and convergence, which will be discussed in later sections.

4.1 Ordinary Linear Model

The model which we will use as the basis of comparison for our primary candidate models will be the gaussian ordinary linear regression model defined as:

$$Y = X'\beta + \epsilon \quad (1)$$

where Y is the vector of true values, X' is the design matrix, β is a vector of coefficients, and ϵ is the residual vector. Estimating the coefficients of the simple linear regression model is fairly straightforward and can be solved in closed form using the method of least squares, as described by Abraham and Ledolter (2006), such that:

$$\hat{\beta} = (X'X)^{-1}X'y \quad (2)$$

4.2 Log-Normal Model

The log-normal regression model is an extension of the ordinary linear regression model and can also be solved in closed form using the method previously proposed. However, the log-normal model involves taking a logarithmic transformation of the response variable before fitting using least squares such that the model is defined as:

$$\log(Y) = X'\beta + \epsilon \quad (3)$$

To interpret the coefficients and predictions, one must apply the inverse of the transformation link by taking the exponentiation of the predicted output or coefficient.

4.3 Gamma Generalized Linear Model

Our final candidate model, and likely the most well suited for our purposes, is the gamma generalized linear model, largely due to its flexible ability to model the mean-variance relationship of the data. The regression equation for the gamma-GLM is displayed in the same form as the log-normal model, both containing a log-link transformation of the response, but with differences in key attributes of the distribution family. Using the definition originally given by Nelder and Wedderburn (1972), with the notation utilized by McCullagh and Nelder (1989), we can write a member of the exponential family as:

$$f(y; \theta) \triangleq \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} - c(y, \phi)\right\} \quad (4)$$

where θ is the natural/canonical parameter and $a(\phi)$ is the dispersion parameter. Utilizing this form provides convenient derivation of $\mu = E(y) = b(\theta)$ and $Var(y) = a(\phi)\nu(\mu) = a(\phi)\dot{b}(\theta)$. These definitions can be further used to fit the generalized linear model by using the Iterative Re-weighted Least Square method to maximize the joint log-likelihood. Using the adaptive Newton-Raphson/Fisher Scoring method, $\hat{\beta}$ can be derived iteratively by:

$$\hat{\beta}^{(j+1)} = \hat{\beta}^{(j)} - [E\ddot{h}(\hat{\beta}^{(j)})]^{-1}\dot{h}(\hat{\beta}^{(j)}) \quad (5)$$

where $E\ddot{h}(\hat{\beta}^{(j)})$ is equivalent to the Fisher information matrix. This equation can also be written in matrix form as:

$$\hat{\beta}^{(j+1)} = (X'\hat{W}^{(j)}X)^{-1}(X'\hat{W}^{(j)}\hat{z}^{(j)}) \quad (6)$$

where $\hat{z}$ is the working response of the previous iteration and $\hat{W}$ is the diagonal matrix of weights with entries $\hat{w}_i = [\nu(\hat{\mu}_i)\dot{\eta}^2(\hat{\mu}_i)]^{-1}$ for link function $\eta(\mu_i)$.

5 Model Selection

Model selection will primarily be two-fold. First, and arguably most importantly, we will validate the model assumptions required for stable inference. Second, we will look at quantitative measures of model accuracy to see if there is separation between models that both meet variance assumptions.

5.1 Residual Diagnostics

By examining the plotted residuals against the fitted values of each model, we are able to assess patterns in the mean-variance relationship that we noted earlier. To support our model assumption of appropriate variance structure, we want to see a residual plot that has no distinct pattern. For analysis, we will utilize Pearson residuals for our standard regression models (defined as $\frac{y_i - \hat{\mu}_i}{\sqrt{\nu(\hat{\mu}_i)}}$) and the analogous deviance residuals (defined as $sign(y_i - \hat{\mu}_i)\sqrt{2[-2\log(\frac{y_i}{\hat{\mu}_i}) + \frac{y_i - \hat{\mu}_i}{\hat{\mu}_i}]}$ for the Gamma generalized linear model).

As can be seen in figure 5, our ordinary linear regression seems to demonstrate a gross violation of accurate variance

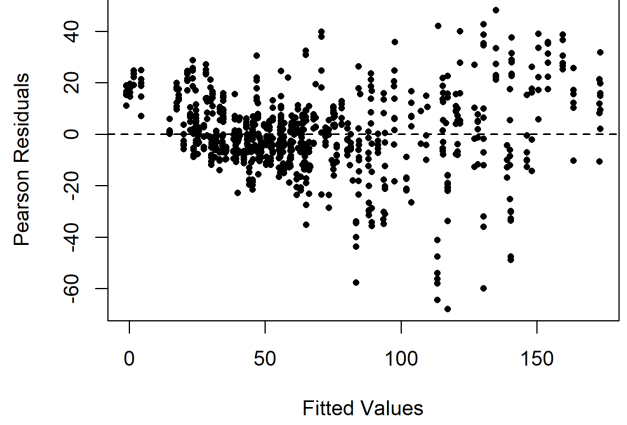
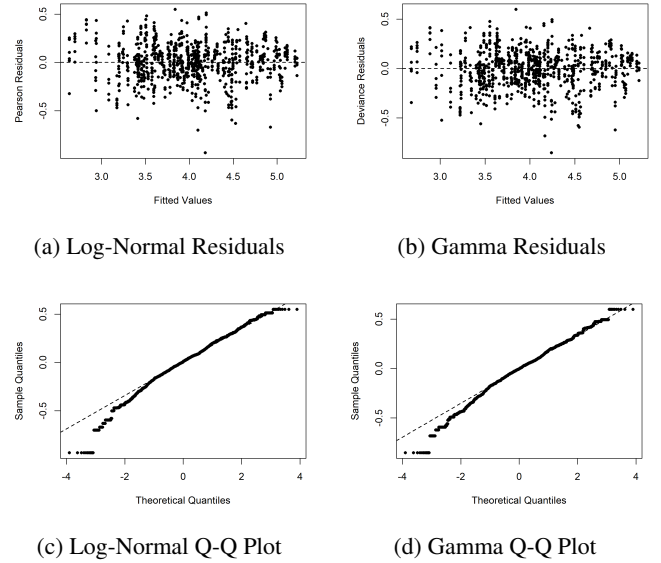


Figure 5: Residuals of fitted ordinary linear regression model plotted against the respective fitted values.

function demonstrated by the funneling outward of residuals with increase in mean. This is to be somewhat expected based on our preliminary analysis of the mean-variance relationship. This issue may be sufficiently addressed by utilizing a log-transformation of the response, but a dynamic variance function scaled by the mean may help to further improve our model fit. We will explore this in the diagnostics of our other candidate models.



We can see from the fitted vs. residuals plots for the Log-Normal and Gamma models in figure 6 that the majority of the original structure among the residuals is removed. In addition, one could potentially argue that the Gamma GLM shows a more even distribution of positive and negative residuals while the log-normal has a higher tendency to underestimate values, but this point is largely subjective.

The Q-Q plots for both models seem to support our analysis of the residuals by demonstrating a strong adherence to the

Q-Q normal line with some deviation in the lower quantiles, likely due to some difficulty that the model has in predicting zero or near-zero yield values. In addition, there may be a slight improvement visible in the residuals of extreme yield values in the Gamma model over the Log-Normal model

5.2 Quantitative Comparison

Beyond the more subjective visual analysis of residuals, there are also some objective measures that can help us validate our assumptions and compare the strength of each model to fitting the data.

Table 1: Quantitative Model Measures

Model	GOF	AIC	BIC
Linear Model	1	86,239.06	86,325.93
Log-Normal Model	1	78,595.52	78,682.39
Gamma Model	1	78,177.51	78,264.38

We can see that there is no significant evidence to say any of the models lack goodness of fit for the data ($p \approx 1$). There is a significant difference in AIC and BIC measures, which quantify how well each model maximizes the likelihood of the response distribution while maintaining model simplicity. We can see that there is a clear favor for the Gamma GLM in this regard, further supporting our analysis that it is the best model to proceed with. We will move forward with this assumption for the rest of our statistical analysis.

6 Final Model

Now that we have selected the best model for our analysis, the primary objectives of our study can be accomplished, that is to evaluate and quantify the relationships between chemical inputs and land management techniques and the yields of the annual row crops they are applied to.

6.1 Summary Statistics

We will begin by examining the summary statistics associated with each variable to get a better understanding of the data we are fitting.

Table 2: Yield Summary by Crop (bu/ac)

crop	n	Mean	SD	Min	Max
Corn	3136	50.17	18.97	11.5	91.7
Soybean	3741	42.03	15.39	10.1	73.4
Wheat	3416	108.82	44.83	25.7	205.0

We can see from our summary statistics that Corn and Soybeans have similar average yields, with Corn being slightly higher than Soybeans but still well within a magnitude of each other. On the contrary, Wheat leads to significantly higher

yields, more than doubling the average yield of both Corn and Soybeans.

Table 3: Summary Statistics for Predictor Variables (kg/ha)

Predictor	Mean	SD	Median	Min	Max
N Fertilizer	14.68	26.91	1.00	0	168.00
P Fertilizer	15.28	27.76	0.00	0	114.00
K Fertilizer	25.79	50.75	0.00	0	377.00
Herbicide Rate	2.56	3.58	0.72	0	25.97

Upon examining our predictor variables, we note that the different variables contain values that are magnitudes, in some cases many, apart from each other. In addition, analysis of the mean, median and maximum values indicate that there may be some very large values that skew the averages. We may deduce that these characteristics may lead to unstable inference and convergence issues when moving forward with model fitting. To address this issue, we will scale the variables before fitting the model. Scaling information will be stored in order to translate model coefficients back into the natural scale when performing inference.

6.2 Fitted Model

Fitting the final model revealed some key issues that were first predicted in early exploratory analysis of the data, mainly that of model convergence. Due to the complex nature of mixed-effects models, parameter estimation via likelihood gradients became unstable, forcing us to reduce the models interaction terms in order to gain stable inference conditions. In determining the variable interactions to drop, those that were known to be the most significant based on previous biological knowledge were favored and therefore removed last in the selection order. Further notes on the convergence issues will be noted in section 10. Model convergence was eventually achieved while still retaining some of the most important interaction effects.

Table 4: Likelihood Ratio Tests for Fixed Effects in the Final Mixed Model

Term Removed	df	AIC	LRT	p-value
None	–	77,549	–	–
Herbicide Rate	1	77,896	349.34	$< 2.2 \times 10^{-16}$
Treatment	3	79,066	1,523.41	$< 2.2 \times 10^{-16}$
Crop $\times$ K Rate	2	77,715	170.38	$< 2.2 \times 10^{-16}$
Crop $\times$ P Rate	2	77,609	64.54	9.646×10^{-15}
Crop $\times$ N Rate	2	77,745	200.19	$< 2.2 \times 10^{-16}$

Since we are utilizing a generalized linear model for fitting our data, we will test parameter significance using the likelihood ratio test proposed by McCullagh and Nelder (1989) rather than the standard Wald’s test more commonly used in ordinary linear regression. The results of the likelihood ratio (Table 4) show that all main and interaction effects in the

model were deemed significant and therefore will be incorporated in the final model listed below.

$$\log(\mu_i) = \mathbf{X}'\beta + \mathbf{Z}u + \epsilon \quad (7)$$

where $\mathbf{Z}$ is the $n \times q$ design matrix and u is the $q \times 1$ vector of coefficients for the random effect. The fitted coefficients for both the random and fixed effects is shown in table 5.

Random Effects

Group	Effect	Variance	Std. Dev.
Year	Intercept	0.006908	0.08311
Residual	–	0.033715	0.18362

Fixed Effects

Effect	Rescaled Estimate	p-value
Intercept	3.77268	$< 2 \times 10^{-16}$
Treatment T2	0.08026	$< 2 \times 10^{-16}$
Treatment T3	0.05151	$< 2 \times 10^{-16}$
Treatment T4	-0.20920	$< 2 \times 10^{-16}$
Soybean	0.021640	0.42593
Wheat	0.892049	$< 2 \times 10^{-16}$
Herbicide Rate	0.0147	$< 2 \times 10^{-16}$
K Rate	0.00604	$< 2 \times 10^{-16}$
P Rate	0.00256	$< 2 \times 10^{-16}$
N Rate	0.00427	$< 2 \times 10^{-16}$
Soybean $\times$ K Rate	-0.00557	$< 2 \times 10^{-16}$
Wheat $\times$ K Rate	-0.00272	7.47×10^{-5}
Soybean $\times$ P Rate	0.00203	6.21×10^{-10}
Wheat $\times$ P Rate	-0.00221	0.00157
Soybean $\times$ N Rate	-0.00345	1.12×10^{-8}
Wheat $\times$ N Rate	-0.00409	$< 2 \times 10^{-16}$

Table 5: Fitted Model Coefficients (rescaled to log scale of original units)

The fitted model provides us with estimates of the vector effect for each interaction or main effect tested. Baseline variables for factored effects are Corn as the base crop effect and T1 as the base treatment effect. The coefficients of the fitted model cannot be directly interpreted as everything has already been converted from the scaled form to the fitted log scale.

7 Results

Now that we have obtained a fitted model, we can move forward with model inference.

7.1 Interpretation - Land Management

Our analysis of land management treatment includes only the main effect of the treatment on crop yield. Relative to conventional land management treatment, no-till land management showed the highest respective yield increase at 8.4%, followed

by the reduced chemical input treatment with a yield increase of 5.3%. The biologically based treatment saw a significant decrease in yield of the magnitude 18.9%.

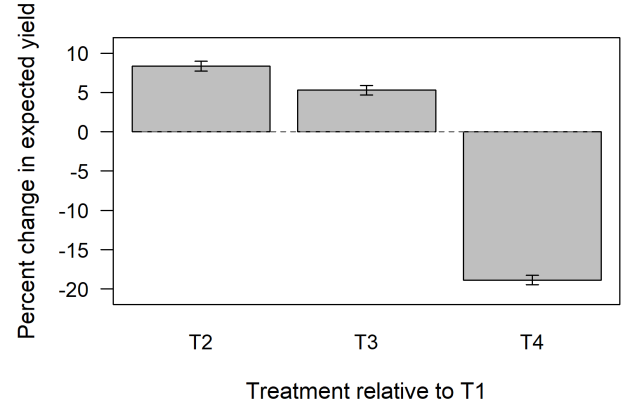


Figure 7: Effect of land management treatment relative to T1 (Conventional Land Management)

7.2 Interpretation - Chemical Inputs

The interpretation of chemical input effects are somewhat more nuanced, especially for that of fertilizer application rates. Before addressing those results, we will quickly take a look at herbicide application rates and their influence on yield. For each kg/ha increase in herbicide application, a 1.5% increase in yield was witnessed. With a mean application rate of 4.429 kg/ha in treatments T1 and T2 (normal chemical inputs conditions), standard application of herbicide can be expected to increase yield by over 6%.

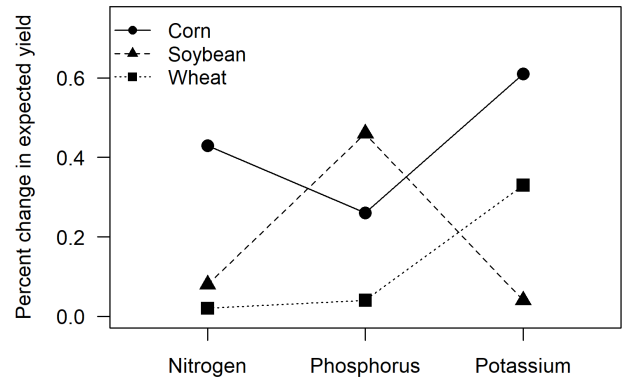


Figure 8: Expected yield increase with use of fertilizer nutrient (kg/ha) by crop species

Fertilizer application rates showed smaller increases for each unit applied, but with the caveat of larger average doses

being utilized. As could be supposed by the nature of different crop species, the impact on yield may be highly dependent on each crop's nutrient dependency on a given fertilizer constituent. We can see from the crop dependent increase for each nutrient, each crop has a specific fertilizer response profile. Corn is highly responsive to Nitrogen and Potassium and moderately dependent on Phosphorus. Soybeans and Wheat can be seen to be much more dependent on a single fertilizer nutrient, Phosphorus for Soybeans and Potassium for Wheat. The specific increases, both on the unit scale and scale of common application can be seen in Table 6. For a more comprehensive examination of recommended fertilizer dosages and their relationship to ag. economics/crop markets, see the recommendations by Franzen and the NDSU Extension Team (2023).

Table 6: Chemical Input Effect on Yield

Chemical Input	Unit Increase	Std. Application
Herbicide	1.48%	6.55%
Nitrogen (Corn)	0.43%	11.27%
Phosphorus (Corn)	0.26%	5.67%
Potassium (Corn)	0.61%	9.96%
Nitrogen (Soybean)	0.08%	0.42%
Phosphorus (Soybean)	0.46%	9.46%
Potassium (Soybean)	0.04%	1.50%
Nitrogen (Wheat)	0.02%	1.03%
Phosphorus (Wheat)	0.04%	0.39%
Potassium (Wheat)	0.33%	9.76%

8 Discussion

The significance of a study only goes so far as the implications which it provides. Our study provides these implications for both land management and chemical input. In the case of land management, no-till methods appear to increase the yield return of a given field. However, as with any non-commercial, localized study, there are practical limitations to the inference. No-till management requires more intensive and time consuming care, as well as producing less consistent yield values. In the MCSE study, yield variance in no-till plots was found to be notably higher than that of its conventionally plowed counterparts with standard deviations of 47.78 and 42.03 bu/ac respectively. In the case of small research operations with no need for consistent returns, this may be feasible. For larger operations, or those in areas with food security issues, the time and risk involved may not be worth the potential high yield (for a meta-analysis on the effects of tillage under various conditions, see Pittelkow et al. 2015).

Chemical inputs inference is possibly more transferrable, but still with some important caveats. While our model demonstrated linear relationships between both herbicide and fertilizer treatments, the assumption of linearity likely holds much stronger for herbicide. In our exploratory analysis, it appears that moderate fertilizer treatments of each of the nutrients resulted in the best yield, with decreasing yields corre-

sponding to further increase. This is likely due to the specific micro and macro-nutrient needs of each crop species [14]. It should also be noted that some of the effects of land management and chemical input may be highly correlated as certain land management treatments were characterized by a lack of chemical input, namely T4.

9 Limitations

There were a few primary limitations that arose during our analysis. They are listed below:

1. Land management techniques are likely correlated to chemical input treatments, potentially misconstruing the true yield effects of each.
2. Model convergence issues forced us to simplify our model, removing possible important interaction effects with each given crop type.
3. Issues of model complexity also forced us to overlook the random effect of replicate plot. This oversight is likely not overly significant as yield variance between replicate plots was shown to be low, but it should be noted regardless.

10 Conclusion

The use of different land management and chemical input treatments demonstrates significant differences in the yield of annual row crops. They are listed below.

1. No-till land management demonstrates consistently higher yields in a localized study at the tradeoff of higher yield variability.
2. Herbicide application proved to be significant and highly effective in improved yield values across all crop types.
3. Fertilizer application had varying effects on yield, though all positive, when conditioned on crop species. Specific nutrient profiles were able to be obtained by examining the effect of each nutrient on a given crop.

11 Extensions - Bayesian Modeling

One of the most significant limitations of our analysis arises from the convergence issues that were associated with fitting more complex mixed-effect models. Based on the recommendation to explore a Bayesian method approach to fitting the model, we may try to perform future analysis of the data using Bayesian posterior models by fitting posterior density functions through the use of Bayes law, resulting in:

$$p(\theta|y) \propto p(\theta)p(y|\theta) \quad (8)$$

where $p(\theta|y)$ is the posterior density function, $p(\theta)$ is the prior distribution and $p(y|\theta)$ is the sampling distribution. This method of analysis is a possible solution for addressing model convergence issues, but goes beyond the scope of this paper.

References

- [1] Daszkiewicz, T. “Food Production in the Context of Global Developmental Challenges,” *Agriculture*, vol. 12, no. 6, p. 832, 2022. doi: 10.3390/agriculture12060832.
- [2] Cook, R. L. and Trlica, A. “Tillage and Fertilizer Effects on Crop Yield and Soil Properties over 45 Years in Southern Illinois,” *Agronomy Journal*, vol. 108, no. 1, pp. 415–426, 2016. doi: 10.2134/agronj2015.0397.
- [3] Khan, I. A., Hassan, G., Malik, N., Khan, R., Khan, H., and Khan, S. A. “Effect of Herbicides on Yield and Yield Components of Hybrid Maize (*Zea mays*),” *Planta Daninha*, vol. 34, no. 4, pp. 729–736, 2016. doi: 10.1590/S0100-83582016340400013.
- [4] Rosenblad, A. “Julian J. Faraway (2016). *Extending the Linear Model with R: Generalized Linear, Mixed Effects and Nonparametric Regression Models*. 2nd ed. Boca Raton, FL: CRC Press. 399 + xiii pp. \$79.96. ISBN: 978-1-4987-2096-0,” *Psychometrika*, vol. 83, no. 4, pp. 1011–1013, 2018. doi: 10.1007/s11336-018-9628-7.
- [5] Kellogg Biological Station Long Term Ecological Research Program (KBS LTER). “KBS LTER,” Michigan State University. Available: <https://lter.kbs.msu.edu/> Accessed: May 9, 2026.
- [6] Haider, S. M. S., Karim, M. M., Ahmed, M. I., Shaheb, M. R., and Shaheenuzzaman, M. “Efficacy of Different Herbicides on the Yield and Yield Components of Maize,” *International Journal of Sustainable Crop Production*, vol. 4, no. 2, pp. 14–16, 2009.
- [7] Noulas, C., Torabian, S., and Qin, R. “Crop Nutrient Requirements and Advanced Fertilizer Management Strategies,” *Agronomy*, vol. 13, no. 8, p. 2017, 2023. doi: 10.3390/agronomy13082017.
- [8] Galton, F. “Regression Towards Mediocrity in Hereditary Stature,” *The Journal of the Anthropological Institute of Great Britain and Ireland*, vol. 15, pp. 246–263, 1886. doi: 10.2307/2841583.
- [9] Abraham, B. and Ledolter, J. *Introduction to Regression Modeling*. Belmont, CA: Thomson Brooks/Cole, 2005.
- [10] Nelder, J. A. and Wedderburn, R. W. M. “Generalized Linear Models,” *Journal of the Royal Statistical Society: Series A (General)*, vol. 135, no. 3, pp. 370–384, 1972. doi: 10.2307/2344614.
- [11] McCullagh, P. and Nelder, J. A. *Generalized Linear Models*. 2nd ed., Chapman and Hall, London, 1989. doi: 10.1007/978-1-4899-3242-6.
- [12] Franzen, D. W. *North Dakota Fertilizer Recommendation Tables and Equations*. Revised 2023, North Dakota State Extension Service, Fargo, 2023.
- [13] Pittelkow, C. M., Linquist, B. A., Lundy, M. E., Liang, X., van Groenigen, K. J., Lee, J., van Gestel, N., Six, J., Venterea, R. T., and van Kessel, C. “When Does No-Till Yield More? A Global Meta-Analysis,” *Field Crops Research*, vol. 183, pp. 156–168, 2015. doi: 10.1016/j.fcr.2015.07.020.
- [14] Mikos-Szymańska, M., Borowik, M., Wyzińska, M., and Rusek, P. “Effects of Different Fertilizer Treatments on Grain Yield and Yield Components of Spring Wheat,” *Research for Rural Development*, vol. 2, pp. 100–106, 2018.